RUIKE LYU

+1-609-436-8628 | rl8728@princeton.edu | https://rick10119.github.io
Princeton, NJ, 08540, USA

Education

- **MAE & Andlinger Center for Energy and the Environment, Princeton University** *Mar. 2025 - Feb. 2026*
  Visiting Student Research Collaborator, Advisor: Prof. Jesse Jenkins Princeton, NJ
- **Department of Electrical Engineering, Tsinghua University** *Sept. 2021 - Jun. 2026*
  Ph.D. in Electrical Engineering, Advisor: Prof. Chongqing Kang Beijing, China
- **Department of Electrical Engineering, Tsinghua University** *Sept. 2017 - Jun. 2021*
  B.E. in Electrical Engineering Beijing, China
- **School of Economics and Management, Tsinghua University** *Sept. 2018 - Jun. 2021*
  Bachelor of Business Administration (Second Degree) Beijing, China

Teaching Experience

- **Department of Electrical Engineering, Tsinghua University** *2023/2024 Academic Year*
  Teaching Assistant for Energy Internet Operation Scheduling and Planning Beijing, China

Selected Publications

J=Journal, C=Conference

English Journal Papers

[J.1] **R. Lyu**, A. Li, J. Wang, H. Luo, E. Du, H. Guo, C. Kang, and J. Jenkins, "Industrial overcapacity can enable seasonal flexibility in electricity use," *Nature Energy*, in press.

[J.2] **R. Lyu**, X. Su, E. Du, H. Guo, Q. Chen and C. Kang, "Efficient Scheduling of Discrete Industrial Processes through Continuous Modeling," *IEEE Transactions on Smart Grid*, vol. 16, no. 6 (2025): 4726-4740.

[J.3] **R. Lyu**, H. Guo, G. Strbac and C. Kang, "Data-Driven Dimension Reduction for Industrial Load Modeling Using Inverse Optimization," *IEEE Transactions on Smart Grid*, vol. 16, no. 3 (2025): 2695-2698.

[J.4] **R. Lyu**, H. Guo, Q. Tang, Q. Chen, and C. Kang, "Production Scheduling Identification: An Inverse Optimization Approach for Industrial Load Modeling Using Smart Meter Data," *IEEE Transactions on Smart Grid*, vol. 16, no. 2 (2025): 1207-1220.

[J.5] Q. Chen, **R. Lyu**, H. Guo, and X. Su, "Real-Time Operation Strategy of Virtual Power Plants With Optimal Power Disaggregation Among Heterogeneous Resources," *Applied Energy*, vol. 361 (2024): 122876.

[J.6] **R. Lyu**, H. Guo, K. Zheng, M. Sun, and Q. Chen, "Co-Optimizing Bidding and Power Allocation of an EV Aggregator Providing Real-Time Frequency Regulation Service," *IEEE Transactions on Smart Grid*, vol. 14, no. 6 (2023): 4594-4606.

[J.7] **R. Lyu**, Y. Gu, and Q. Chen, "Electric Vehicle Charging Right Trading: Concept, Mechanism, and Methodology," *IEEE Transactions on Smart Grid*, vol. 13, no. 4 (2022): 3094-3105.

[J.8] Q. Chen, X. Fang, H. Guo, K. Zheng, Q. Tang, **R. Lyu**, K. Pan, P. Palensky, D. S. Kirschen, and C. Kang, "The Competition and Equilibrium in Power Markets Under Decarbonization And Decentralization," *iEnergy*, vol. 1, no. 2 (2022): 188-203.

Conference Papers

[C.1] **R. Lyu**, H. Guo and Q. Chen, "Approximating Energy-Regulation Feasible Regions of Virtual Power Plants: A Data-Driven Inverse Optimization Approach," *2024 IEEE Power & Energy Society General Meeting (PESGM)*, Seattle, WA, USA, 2024, pp. 1-5.

[C.2] **R. Lyu**, H. Guo, Y. Zheng, Y. Bai and Q. Chen, "LSTN: A Linear Model of Industrial Production Process for Demand Response," *2023 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE)*, Grenoble, France, 2023, pp. 1-5.

[C.3] J. Li, **R. Lyu**, Y. Zhang, X. Cha, K. Zheng, and H. Guo, "Value-Based Industrial Load Shedding with Supply Chain Coordination Reduces Electricity Cost," *2026 IEEE PES International Meeting*, in press.

[C.4] J. Zhang, **R. Lyu**, X. You, J. Wang, Y. Cai, and H. Guo, "When Will Real-Time Pricing Outperform Time-of-Use Pricing Significantly?," *2026 IEEE PES International Meeting*, in press.

[C.5] H. Huang, **R. Lyu**, C. Feng, H. Zhong, H. B. Gooi, B. Li and R. Liang, "Learning for Feasible Region on Coal Mine Virtual Power Plants with Imperfect Information," *2025 IEEE PES General Meeting*, in press.

[C.6] Q. Liu, **R. Lyu**, Z. Zhai, Y. Shen, X. Liu and H. Guo, "Integrating Fast-response Capability into Virtual Power Plant Operation for Ancillary Services," *2025 IEEE Powertech*, Kiel, Germany.

[C.7] R. Chen, Z. Tang, **R. Lyu**, Q. Zheng, H. Song and H. Guo, "Combining AI and Simulation to Assess Building Demand Response Potential at Scale," *2025 5th International Conference on Advances in Electrical, Electronics and Computing Technology (EECT)*, Guangzhou, China, 2025, pp. 1-6. (**Best Presentation**)

[C.8] Y. Shen, **R. Lyu**, H. Guo and C. Kang, "An Improved Modeling Method for Electrolyte Aluminum Loads Considering Thermal Balance and Flexible Regulation Cost," *2025 IEEE International Conference on Power Systems and Smart Grid Technologies (PSSGT)*, Chongqing, China, 2025, pp. 331-337. (**Best Presentation**)

[C.9] A. Luo, **R. Lyu**, H. Guo, Y. Cai and Q. Chen, "An Incentive-Compatible VPP Profit Allocation Model Considering the Operating Characteristics of Air Conditioning Loads," *2024 7th International Conference on Energy, Electrical and Power Engineering (CEEPE)*, Yangzhou, China, 2024, pp. 1270-1278. (**Best Presentation**)

[C.10] L. Su, **R. Lyu**, H. Guo, Y. Cai and Q. Chen, "Coordinating Air Conditioning Load Clusters to Provide Frequency Regulation Using Setpoint Changes," *2024 7th International Conference on Energy, Electrical and Power Engineering (CEEPE)*, Yangzhou, China, 2024, pp. 905-910.

Chinese Journal Papers

[J.8] X. Su, **R. Lyu**, Y. Bai, X. Wang, W. Zhao, and H. Guo, "A Method for Modeling the Feasible Region of Industrial Production Processes Based on the Optimal Adjustable Load Model," *Automation of Electric Power Systems*, in press.

[J.9] X. Su, **R. Lyu**, H. Guo, and Q. Chen, "A Method for Optimal Selection of High-Capacity Industrial Users for Demand Response Based on Load Step Data Processing Mode," *Electric Power*, 2024, 57(01): 18-29.

[J.10] Q. Chen, **R. Lyu**, H. Guo, H. Jia, Y. Ding, Y. Wang, and C. Kang, "Electricity User Behavior Modeling for Demand Response: Research Status Quo and Applications," *Electric Power Automation Equipment*, 2023, 43 (10): 23-37.

[J.11] Q. Chen, **R. Lyu**, Q. Tang, K. Li, H. Gao, and H. Guo, "Emergency Response of Electricity Market from Perspectives of Suspension of Spot Market in Australia," *Automation of Electric Power Systems*, 2022, 46 (16): 214-223.

PATENTS

[P.1] Q. Chen, Y. Chen, K. Zheng, and **R. Lyu**, "Method for estimating state of power based on electrochemical model of lithium-ion battery," US Patent App. 18/919,593, 2025.

[P.2] Q. Chen, Y. Chen, K. Zheng, and **R. Lyu**, "Method for updating state of charge based on power characteristic of electrochemical model of lithium-ion battery," US Patent App. 18/919,638, 2025.

HONORS AND AWARDS

- **Best Presentation Award**, IEEE PES Ph.D. Dissertation Challenge *2025*
- **Best Presentation/Paper Award**, PESGM2025, EECT2025, PSSGT2025, CEEPE2024
- **Outstanding Undergraduate Student Mentor**, Tsinghua University *2024*
- **National Scholarship for Graduate Students**, Ministry of Education of China *2023, 2025*
- **First-Class Comprehensive Excellence Scholarship**, Tsinghua University *2022*
- **Future Scholar Scholarship**, Tsinghua University *2021*
- **Outstanding Bachelor Thesis**, Tsinghua University *2021*
- **Outstanding Student Leader Award**, Tsinghua University *2020*

PARTICIPATED RESEARCH PROJECTS

- **Key Technologies of Interactive Regulation of Large-Scale Flexible Resources Aggregation in Virtual Power Plant**
  The National Key Research and Development Program of China (2021YFB2401200), 2021 - 2025
- **Energy Conservation and Supply-Demand Interaction Technology for High-Energy-Consumption Industrial Users**
  The National Key Research and Development Program of China (2023YFB2407300), 2023 - Present

PARTICIPATED INDUSTRIAL PROJECTS

- **Optimized Operation Strategy for Auto Manufacturing Plant** *State Power Investment Corporation (SPIC)*
  Chongqing, China 2024
  - This project involved designing an optimized operation strategy for an auto manufacturing plant (BYD) in Chongqing, integrating 60MW industrial load (manufacturing facilities), 60MW/240MWh energy storage system, and 10MW solar generation capacity. The project is generating approximately $4 million in annual savings, demonstrating the significant value that can be created through intelligent energy management of industrial loads and distributed energy resources.

INVITED TALKS

- **Columbia University** *Feb. 9, 2026*
  Host: Prof. Bolun Xu New York, NY
- **Princeton University** *Feb. 6, 2026*
  Host: Prof. Denise Mauzerall Princeton, NJ
- **Harvard University** *Feb. 4, 2026*
  Host: Dr. Chris Nielsen Cambridge, MA
- **Massachusetts Institute of Technology (MIT)** *Feb. 3, 2026*
  Host: Dr. Ruaridh MacDonald Cambridge, MA
- **Dartmouth University** *Feb. 2, 2026*
  Host: Prof. Cong Chen Hanover, NH
- **New York University** *Jan. 28, 2026*
  Host: Prof. Yuzhang Lin New York, NY
- **Stanford University** *Dec. 5, 2025*
  Host: Prof. Adam Brandt Stanford, CA
- **University of Washington** *Oct. 17, 2025*
  Host: Prof. Baosen Zhang Seattle, WA
- **California Institute of Technology (Caltech)** *Oct. 16, 2025*
  Host: Prof. Steven Low Pasadena, CA
- **University of California, Riverside** *Sept. 26, 2025*
  Host: Prof. Nanpeng Yu Riverside, CA

REFERENCES

1. **Prof. Chongqing Kang**
   Professor, Dean of Dept. Electrical Engineering, Tsinghua University
   Email: cqkang@tsinghua.edu.cn
2. **Prof. Jesse Jenkins**
   Associate Professor, MAE & Andlinger Center for Energy and the Environment, Princeton University
   Email: jdj2@princeton.edu
3. **Prof. Daniel Kirschen**
   Professor, Dept. Electrical & Computer Engineering, University of Washington
   Email: kirschen@ece.uw.edu